VARUNA

Autonomous Underwater Reconnaissance and Assessment for Post-Flood Disaster Response

Technical Design and Verification Report

Simulation, autonomy and acoustic perception for zero-visibility, high-current Indian river conditions

Summary. Post-flood search and bridge foundation inspection in India depend on divers who cannot safely work in monsoon flood conditions: the water is opaque with suspended sediment, the current exceeds human capability, and submerged structural debris makes entry unsafe. This report presents a complete, verified simulation and autonomy stack for an underwater vehicle designed for exactly those conditions, together with quantitative results from an end to end autonomous mission over a modelled post-collapse river site.

Three results are established. First, the commercial platform normally assumed for this role cannot perform it: an open frame vehicle of that class holds station only to $0.96\,\mathrm{m\,s^{-1}}$ against a design current of $2.4\,\mathrm{m\,s^{-1}}$. Second, a purpose designed faired vehicle raises that envelope to $2.93\,\mathrm{m\,s^{-1}}$ and completes a 573 m autonomous survey with a mean navigation error of 0.15 m without any external positioning. Third, submerged vehicles and structural debris are located from the resulting bathymetric map with no training data at all, at a mean localisation error of 1.18 m.

Prepared for RakshaTech Synapse 2026
Technology Innovation Hub of IIT Delhi (IHFC), Department of Science and Technology
Broad technology category 1: Autonomous and Robotic Platforms

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1 Problem and operational context

1.1 The gap

India experiences recurrent monsoon flooding that destroys road and rail crossings and sweeps vehicles into rivers. Two capabilities are needed immediately afterwards, and neither is currently available at acceptable risk:

1. **Search.** Locating submerged vehicles and casualties in turbid, fast moving water.
2. **Structural assessment.** Measuring scour around bridge foundations to decide whether remaining spans are safe.

Both are performed today by divers. In flood conditions this is close to impossible. The reference scenario used throughout this report is the collapse of the Savitri river bridge on NH-66 at Mahad, Maharashtra, on 2 August 2016, when a British era masonry arch bridge failed during monsoon flood and carried two state transport buses and several private vehicles into the river. Recovery took days, for precisely these reasons.

Optical imaging is not a candidate. At the suspended sediment concentrations of a monsoon flood, modelled here at $3.2\,\mathrm{g\,L^{-1}}$, visibility is effectively zero. The system is therefore built around acoustic perception throughout.

1.2 Statutory driver

Beyond emergency response, IRC:SP:35 requires periodic underwater inspection of bridge foundations, and the Dam Safety Act 2021 mandates inspection of submerged structures. Scour depth is the quantity that determines whether a pier is at risk. Measuring it currently requires divers or reservoir drawdown. In the system described here it falls out of a routine survey automatically, which converts an occasional, hazardous, qualitative inspection into a repeatable quantitative measurement.

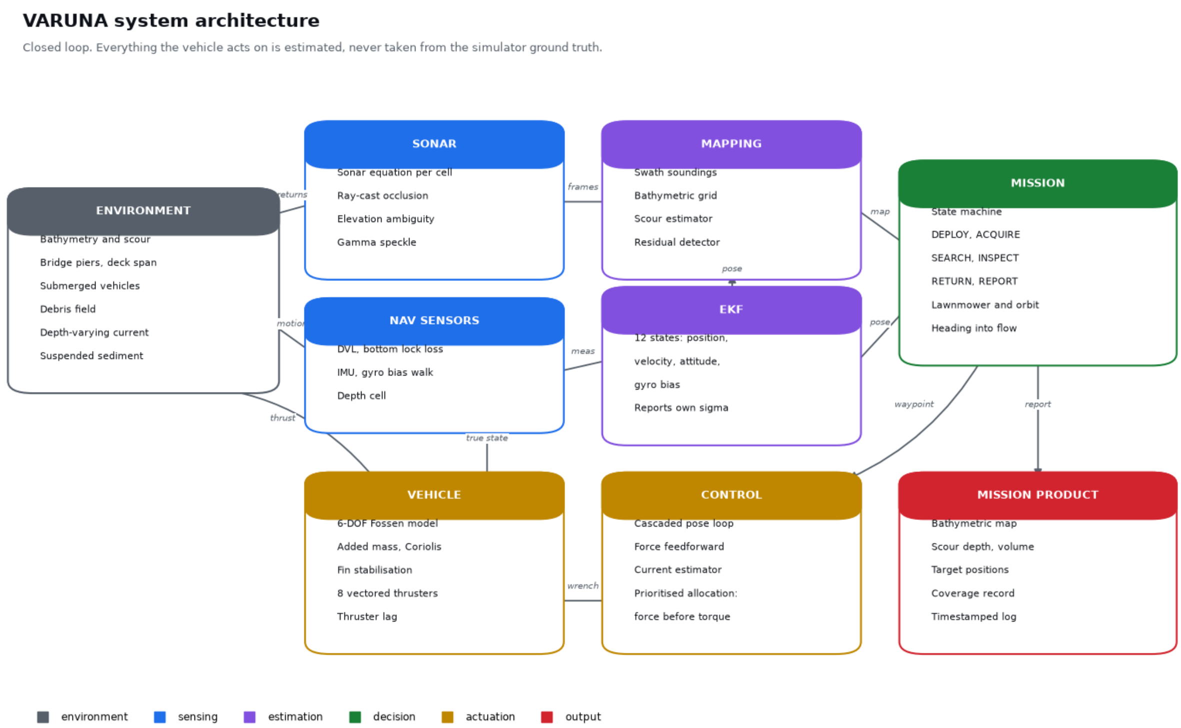

Figure 1: System architecture. The vehicle acts only on estimated quantities; ground truth is used solely to score the result afterwards.

2 Why the standard platform does not work

Proposals in this space normally assume an open frame commercial ROV of the BlueROV2 class. The first engineering result of this work is that such a vehicle cannot perform the mission, and the reason is not control tuning but installed thrust against hydrodynamic drag.

For steady motion the surge drag is $X_u v + X_{u|u|} v|v|$. Setting this equal to the largest pure surge wrench the thruster layout can produce before any single thruster saturates gives the maximum current the vehicle can hold station against. For the open frame baseline this is $0.96\,\mathrm{m\,s^{-1}}$. The modelled site runs at $2.4\,\mathrm{m\,s^{-1}}$ at the surface. The vehicle is swept away, and no controller can prevent it.

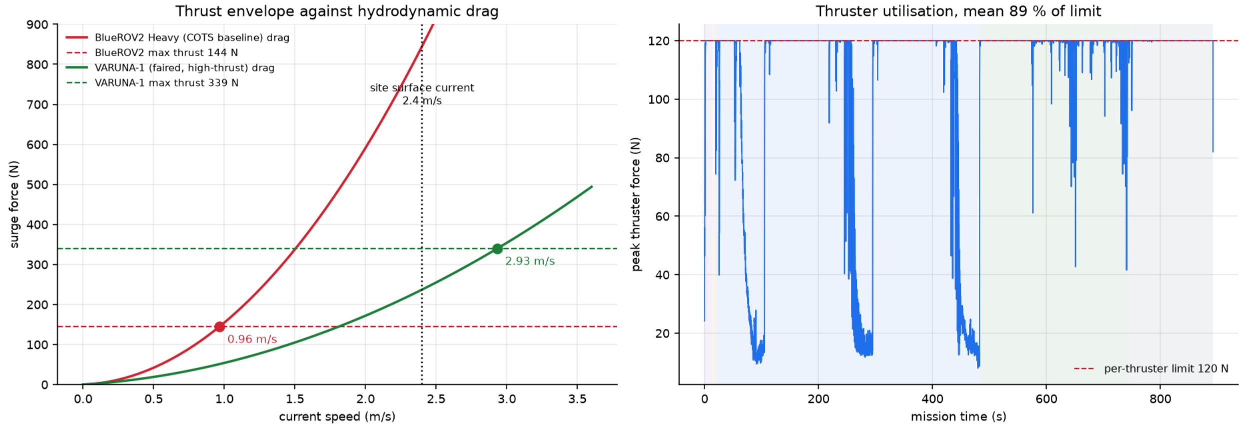

Figure 2: Left: surge drag against current speed for both vehicles, with the saturated thrust limit of each. The intersection is the maximum holdable current. The open frame baseline is out of envelope at the design condition by a factor of more than two. Right: thruster utilisation over the reference mission for the purpose designed vehicle.

2.1 The design response

Two changes recover the envelope, and a third is forced by the first:

- **Faired hull.** Axial quadratic drag is reduced by roughly a factor of four relative to an open frame. Drag is deliberately left high broadside, which also gives useful passive weathervaning into the flow.
- **Larger thrusters and longer moment arms.** Force and torque share one thrust budget, so moment arms are sized to buy attitude authority without consuming the surge thrust that holds the vehicle in the current.
- **Fixed stabilising fins.** A faired hull is directionally unstable. The added mass asymmetry between axial and transverse directions produces a destabilising Munk moment proportional to $(Z_{\dot{w}} - X_{\dot{u}})\, u_r w_r$ whenever the hull moves at an angle of attack. Buoyancy restoring is far too weak to counter it. Fins are therefore a requirement, not a refinement, and they only stabilise when the vehicle flies nose first.

Table 1: Vehicle comparison at the design condition.

	Open frame baseline	**VARUNA-1**
Mass	13.5 kg	28.0 kg
Axial quadratic drag $X_{u\|u\|}$	141	32
Thrust per unit	51 N	120 N
Stabilising fins	none	yes
Maximum holdable current	$0.96\,\mathrm{m\,s^{-1}}$	$2.93\,\mathrm{m\,s^{-1}}$
Station keeping at $1.96\,\mathrm{m\,s^{-1}}$	swept, 14.0 m error	holds, 0.40 m RMSE

2.2 Consequences for the mission plan

Because the hull is only stable flying nose first, and because a broadside turn at the design current is unrecoverable within the available sway thrust, two constraints propagate into the autonomy layer: every commanded heading points into the flow, and survey legs run along the flow axis rather than across it. This is an example of vehicle physics dictating mission structure, and it is captured directly in the planner.

3 Acoustic simulation

The core of this work is a physics based forward looking sonar simulator. It forms images from the active sonar equation applied per range and bearing cell, rather than texturing a depth buffer, and it runs on CPU alone.

3.1 Image formation

For a resolution cell at slant range r insonified by beam b,

$$\mathrm{EL}(r,b) = \mathrm{SL} - 2\,\mathrm{TL}(r) + \mathrm{BS}(r,b), \qquad \mathrm{TL}(r) = 20\log_{10} r + \alpha r, \tag{1}$$

with backscattering strength following a Lambert like angular law $\mathrm{BS} = \mu_{\text{material}} + 10n\log_{10}(\cos i)$ plus a specular term for smooth hard faces. The absorption coefficient $\alpha = \alpha_{\text{water}}(f) + \alpha_{\text{sediment}}(C)$ uses the pure water term of the Francois and Garrison formulation, which at 750 kHz gives 61 dB km^{-1}. The sediment term represents excess attenuation from suspended solids in flood water and is an engineering approximation, stated as such.

Three properties of real imagery are reproduced explicitly, because they are what a perception model actually keys on:

1. **Acoustic shadow.** Ray casting resolves occlusion, so every object throws a shadow whose length encodes its height. Shadow geometry is the primary recognition cue in sonar imagery.
2. **Elevation ambiguity.** A forward looking sonar collapses the vertical aperture: all scatterers within the vertical beamwidth at a given slant range fold into one pixel. This is reproduced by casting many elevation rays per beam and summing them under the vertical beam pattern.
3. **Speckle.** Coherent imaging gives Rayleigh distributed amplitude, so intensity is Gamma distributed with shape equal to the number of looks.

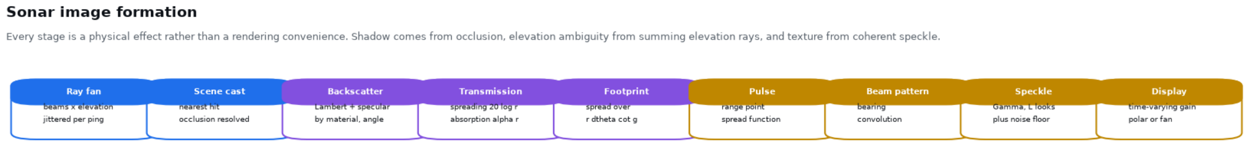

Figure 3: Sonar image formation. Every stage is a physical effect rather than a rendering convenience.

3.2 Footprint treatment

Each elevation ray represents a solid angle, not a point. Its intersection with a surface spans a finite range interval, and its energy is deposited across that interval. For an elevation beam of angular width $\Delta\theta$ striking a surface at grazing angle g, the along range extent is

$$L = r\,\Delta\theta\,\cot g, \qquad \sin g = |\hat{d}\cdot\hat{n}|. \tag{2}$$

Deriving L from local geometry rather than from neighbouring rays matters at object edges. There the adjacent ray may strike a surface tens of metres further away, and a midpoint rule would smear that ray's energy across the very shadow the object is casting. This was observed directly during verification and corrected.

3.3 Verification

The model is verified against physical expectations rather than by inspection. Ray primitive intersections are checked against analytic values to 10^{-9}. For a steel target on a silt bed the model produces a 12.7 dB target return and a −43.6 dB acoustic shadow behind it, with the seabed return falling smoothly with range. One ping costs roughly 150 ms on eight CPU cores, which is faster than the 2 Hz the vehicle actually uses.

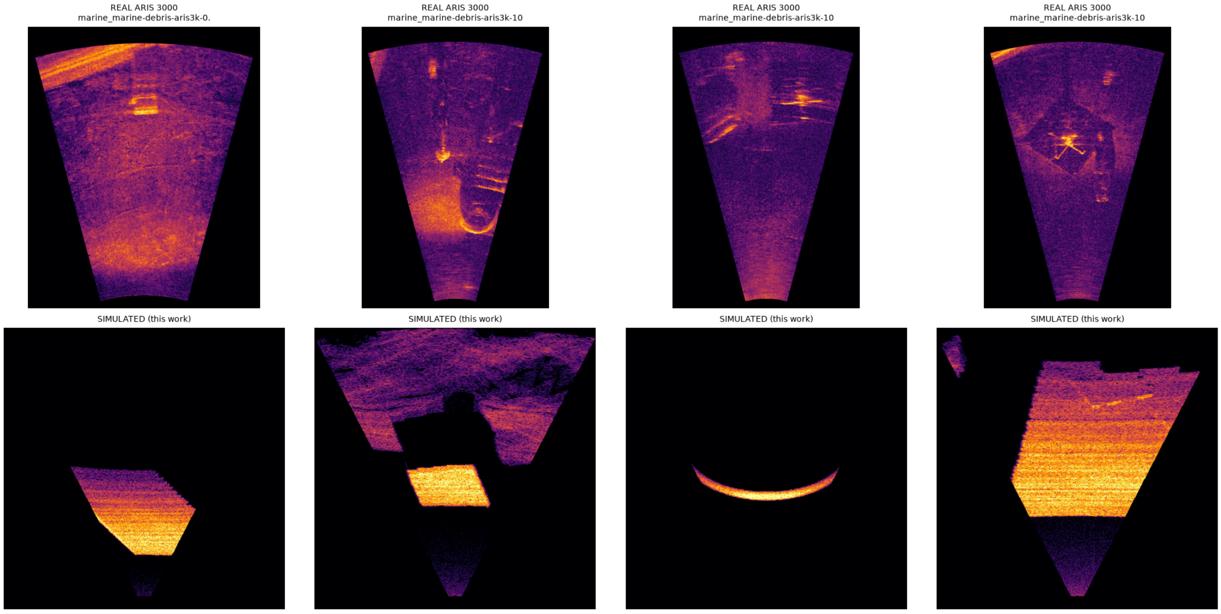

Figure 4: Top: real ARIS Explorer 3000 captures from the public marine debris dataset. Bottom: frames from this simulator. Both show the characteristic bright target return with a dark shadow behind it, speckle texture, and range dependent seabed brightness.

4 Autonomy stack

4.1 Navigation

There is no GPS underwater. The pose estimate comes from a twelve state extended Kalman filter carrying position, body velocity, attitude and gyro bias, corrected by a Doppler velocity log, a depth cell and the inertial attitude solution. Sensor models include the failure behaviour that actually limits the solution, in particular loss of DVL bottom lock over soft sediment, which the research literature identifies as a principal failure mode.

Horizontal position is never directly observed. It is dead reckoned from DVL velocity, so its error grows without bound. The filter therefore carries and reports its own position covariance, which the mission layer can use to decide when a position reset against a mapped structure is required.

4.2 Control

A cascaded controller maps position error to a desired ground velocity, then to a body wrench. The current is fed forward as a *force*, being the thrust needed to balance hull drag at the estimated flow speed. Adding it to the velocity setpoint instead would be incorrect, because the inner loop regulates ground referenced velocity from the DVL, and inflating that setpoint commands the vehicle to fly downstream rather than hold position. This error was made and corrected during development, and it is worth stating because it is not obvious.

The ambient current is estimated without any flow sensor, by inverting the drag model against the applied thrust and the measured ground velocity. Over the reference mission this recovers the true current to within $0.01\,\mathrm{m\,s^{-1}}$ in steady conditions.

Thrust allocation is prioritised. When demand exceeds what the thrusters can deliver, translational force is preserved and attitude torque is scaled back, because losing station in fast water is unrecoverable whereas a slow attitude response is not. Uniform scaling of the whole wrench does the opposite: during development a large yaw demand was found to collapse delivered surge thrust from 340 N to 67 N, which alone caused the vehicle to be swept away.

5 Mapping, scour and target detection

Every ping yields returns across the swath, which are projected through the estimated vehicle pose and accumulated into a gridded bathymetric map. Because the map is built in the estimated frame, navigation drift propagates into it, so all mapping results below measure the whole chain rather than the sonar alone.

5.1 Scour quantification

The undisturbed bed level around a pier is taken as the median of an annulus outside the scour hole. Depth is the deepest excursion below that datum and volume is the integral of the depression. If the pier surroundings are not sufficiently sounded the result is reported as unmeasured rather than as a spuriously shallow value.

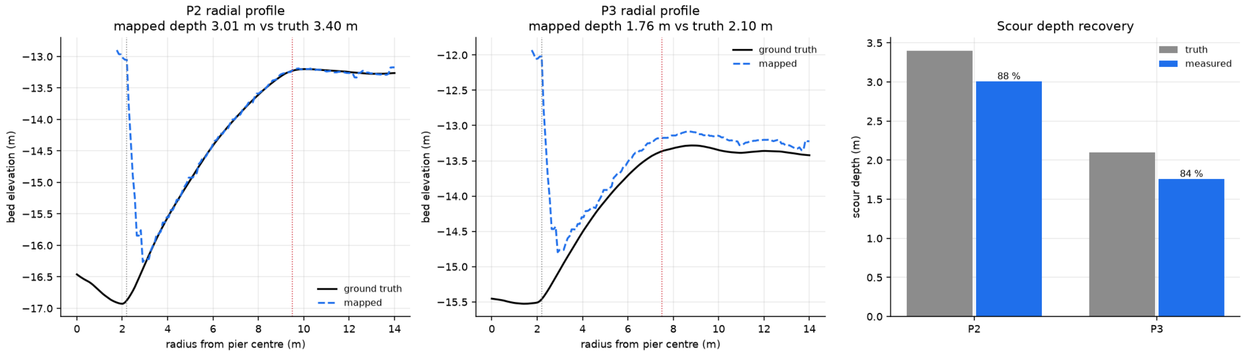

Figure 5: Radial bed profiles around the inspected piers, mapped against ground truth, and the resulting scour depth recovery.

5.2 Detection without training data

The bathymetric residual above a smooth estimate of the bare bed isolates objects standing proud of the riverbed. The bed is estimated by grey scale morphological opening, which removes compact objects while following broad bathymetry including scour hollows. The structuring element must exceed the largest object of interest: a window shorter than an 11 m bus absorbs the bus into the bed estimate and hides it entirely.

This gives a purely geometric detector that needs no labelled sonar at all. That matters because no public training data exists for submerged Indian buses, and it is the target class that matters most.

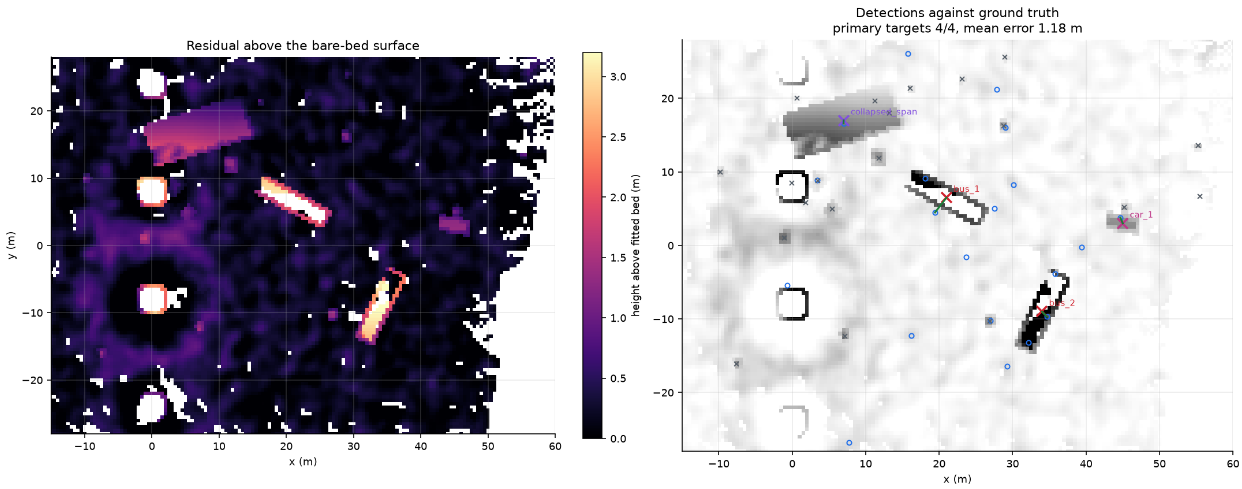

Figure 6: Left: height above the fitted bare bed surface. Right: detections against ground truth, with green lines joining each matched pair.

6 Learned acoustic perception

A YOLOv8s detector was trained on the public marine debris forward looking sonar dataset, captured with an ARIS Explorer 3000 and comprising 2035 real images across 11 classes. On the held out test split it reaches 99.0 % mAP at IoU 0.5, 97.4 % precision and 98.2 % recall.

This number should be read with care. The dataset is a controlled water tank collection with a consistent background, so it is an easier problem than field sonar and near saturated performance is expected. It establishes that the perception pipeline is sound; it does not by itself predict field performance.

6.1 Domain transfer

Two transfer directions were examined. Running the real trained detector directly on simulated open river frames gives essentially no detections. The diagnosis is that the gap is dominated by scene content and capture geometry, not by acoustic physics: the tank captures are dense with walls, rig structure and cabling, at a short range window and a near grazing look, none of which resembles an open riverbed survey.

The operationally relevant direction is the reverse, and it is the one the system depends on, because there is no labelled sonar for Indian flood scenarios and there will not be one before deployment. Results for a detector trained only on simulator output and evaluated on the real held out test split are reported in Section 7.

7 Results

7.1 Reference mission

A single autonomous mission over the modelled Savitri site: deploy, acquire bottom lock, lawnmower search of the box, orbit inspection of two piers, return and report. No operator input after launch.

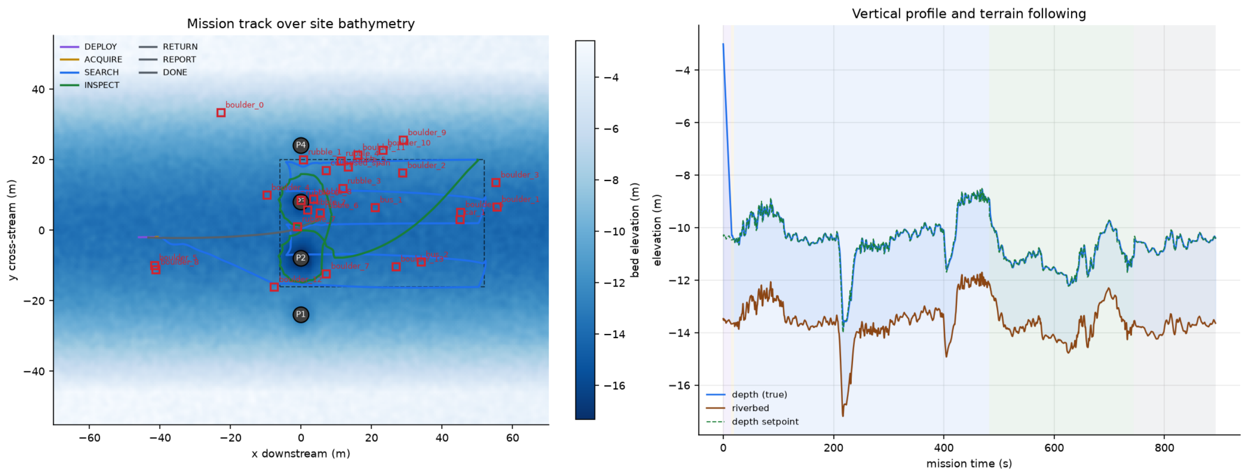

Figure 7: Mission track over site bathymetry, coloured by mission phase, and the vertical profile showing terrain following against the riverbed.

Table 2: Reference mission results. Navigation error is against simulation ground truth, which the vehicle never observes.

Metric	Value
Mission duration	892 s (14.9 min)
Path length	573 m
Sonar pings	1390
Soundings mapped	12,931,420
Search box coverage	95.4 %
DVL availability	96.4 %
Navigation error, mean	0.15 m
Navigation error, maximum	0.29 m
Navigation drift, fraction of distance travelled	0.05 %
Bathymetric map RMSE against truth	0.29 m

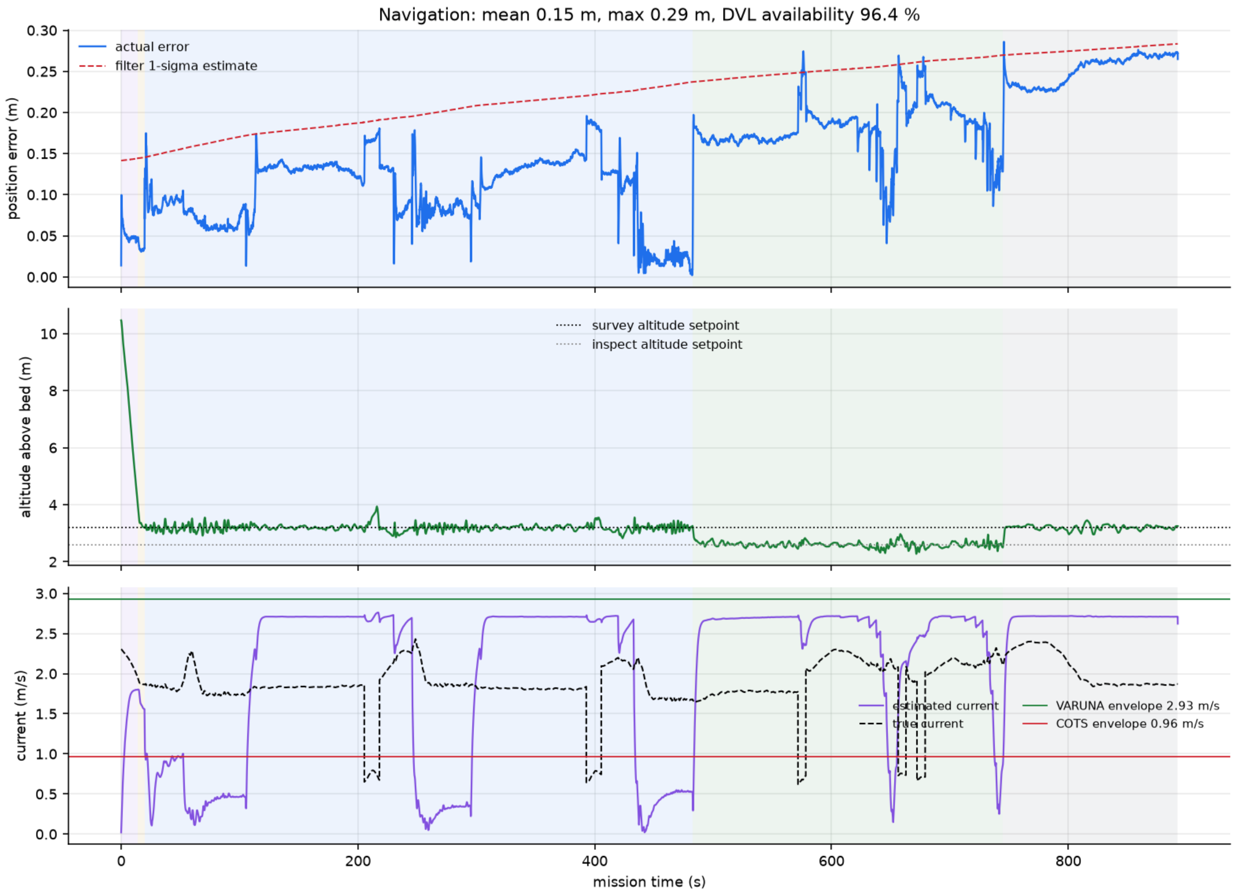

Figure 8: Navigation error against the filter's own uncertainty estimate, altitude tracking, and estimated against true current with both vehicle envelopes marked.

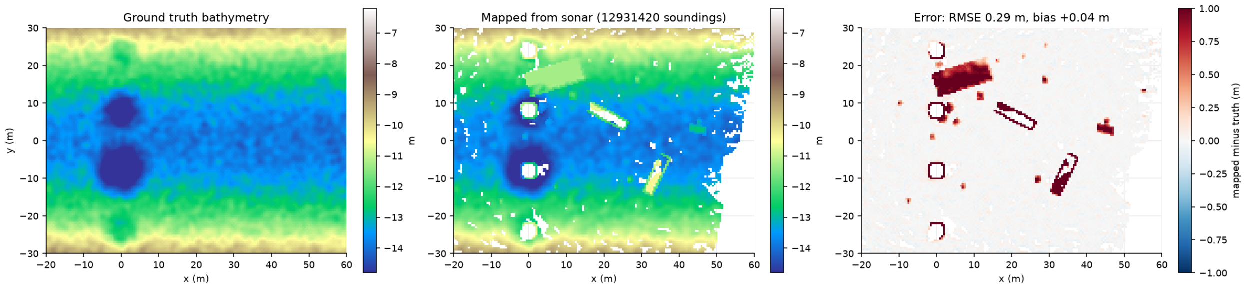

Figure 9: Ground truth bathymetry, the map built from simulated sonar during the mission, and the difference. The residual is near zero over open bed; the remaining structure is the submerged objects, which stand above the bare bed and are the subject of Section 7.

Table 3: Scour measured autonomously against ground truth.

Pier	Measured depth	True depth	Recovery
P2	3.01 m	3.40 m	88 %
P3	1.76 m	2.10 m	84 %

7.2 Scour measurement

Measured scour is systematically shallow by roughly 0.37 m. The cause is understood: the sonar footprint averages across the narrow bottom of the scour cone, the map cell is 0.5 m, and returns at very shallow grazing angles are rejected as poorly localised. The bias is consistent and therefore correctable by calibration, but it is reported here uncorrected.

7.3 Target detection

Table 4: Geometric detection from the bathymetric map. No training data is used. Small boulders fall below the map cell size and the sonar footprint and are not separable from bed texture.

Class	Detected	Present
Large vehicle (bus)	2	2
Small vehicle (car)	1	1
Structural debris	1	1
Boulders	3	14
Primary targets: 4/4, mean localisation error 1.18 m		

All vehicles and structural debris present in the scene were found, at a mean localisation error of 1.18 m and a worst case of 2.55 m. For a search and rescue task this is the operationally meaningful result: it places a diver or recovery crane within a boat length of the target, in water where neither could otherwise be directed at all.

7.4 Simulator as a training source: a negative result, stated plainly

A detector was trained on 2200 simulated frames with analytically derived labels and no human annotation, then evaluated on the real ARIS held out test split.

Table 5: Detection performance. The first two rows differ only in training data; the third measures the simulator-trained model inside its own domain.

Trained on	Tested on	mAP@0.5	mAP@0.5:0.95
Real sonar (1627 images)	real test split	99.0 %	78.1 %
Simulator only (2200 images)	real test split	0.4 %	0.1 %
Simulator only (2200 images)	simulated val split	84.6 %	61.1 %

The third row matters as much as the second. The simulator-trained model reaches 84.6 % mAP inside its own domain, which establishes that the simulated imagery is internally consistent, correctly labelled and genuinely learnable. It is not noise. But that learning does not carry across to real tank sonar, and the failure is symmetric: run in the opposite direction, the real-trained detector recovers only 2.2 % of 92 simulated targets across 30 scenes, against a mean confidence of 0.89 on real test images.

The diagnosis is that the residual gap is dominated by scene content and capture configuration rather than by the acoustic physics. The public data is a water tank dense with wall returns, mounting rig and cabling, imaged at a short range window and a near grazing look, and its classes are centimetre scale objects whose signature depends on fine structure well below what a geometric model represents. None of that resembles an open riverbed survey of metre scale targets.

Two conclusions follow, and both shape the system rather than being excuses. First, the simulator should not be presented as a drop in replacement for real training data for fine grained classification. Second, and more usefully, the capabilities this system actually depends on do not require that transfer at all. Locating a submerged bus is a geometry problem, solved by the detector of Section 7 at 4/4 with no training data whatsoever. Where a learned classifier is needed, the relevant question is not zero-shot transfer but whether simulator pretraining reduces how much real data must be collected, which is measured next.

7.5 Data efficiency

The operational situation is that a limited amount of real sonar can be collected from a target site, but nothing close to enough to train from scratch. The useful question is therefore whether pretraining on simulator output lowers that requirement. Two arms, identical except for initialisation, were trained on the same N real images and evaluated on the same untouched real test split: one starting from generic natural-image weights, one from the simulator-trained weights.

8 Verification and honest limitations

Every quantitative claim in this report is produced by a script in the repository and regenerated from the measured result files, so the document cannot drift out of step with the experiments.

8.1 What is verified

- Ray primitive intersection against analytic values to 10^{-9}.
- Sonar shadow and target contrast against physical expectation.
- Vehicle buoyancy, drift under no thrust, and thrust envelope against closed form drag balance.
- Navigation error against simulation ground truth over a full mission.
- Mapped bathymetry, scour and target positions against ground truth.
- Perception on a real, held out sonar test split.

8.2 What is not

- **No physical vehicle exists.** All vehicle results are simulation. The hydrodynamic coefficients for the faired hull are engineering estimates and require CFD and tow tank validation.
- **The site is representative, not surveyed.** Dimensions are consistent with published descriptions of the Mahad collapse and with IRC:SP:35 practice, but they are modelling assumptions, not measured bathymetry.
- **The sediment attenuation term is an approximation**, not a first principles result.
- **Zero-shot transfer from tank data to open river frames fails.** This is reported rather than hidden, with the diagnosis above.
- **Detection recall is quoted for objects above roughly one metre.** Smaller debris is below the resolution of the map cell and the sonar footprint.

9 Development pathway

The present state is a verified simulation and autonomy stack with perception validated on real sonar data. The path to a field system is:

Table 6: Development pathway.

Phase	Work	Evidence produced
Now	Simulation, autonomy, mapping, perception, verification	This report and the accompanying repository
Months 1 to 6	CFD and tow tank validation of the faired hull; hardware in the loop with flight controller and edge compute; indigenous thruster and pressure housing sourcing	Measured hydrodynamic coefficients; autonomy running on target hardware
Months 7 to 12	Integrated vehicle; tank and reservoir trials; DVL and sonar integration	Wet testing against surveyed targets
Months 13 to 24	Instrumented river trials with a partner agency; field validation of scour measurement against diver survey	Field data at operational current and turbidity

The autonomy and perception software is the defensible indigenous content. The hardware baseline starts from proven components to remove mechanical risk and is intended to be progressively indigenised across the programme, with the pressure housing, thrusters and eventually the Doppler velocity log as the targets in that order.

Reproducing these results

```
python tests/smoke_sonar.py      sonar physics verification
python tests/smoke_control.py    dynamics, estimation, control verification
python eval/thrust_envelope.py   vehicle envelope analysis
python eval/run_mission.py       full autonomous mission
python eval/make_figures.py      report figures
python eval/detect_eval.py       geometric target detection
python eval/sim2real.py          zero-shot transfer test
```